

GRA Photovoltaics: Extended Theory and Technology of Topological Entropy Nullification

Synthesis of meta-solar cells with a 1000-fold increase in specific energy productivity (NHI) from widely available materials

qqewq Ecosystem
<https://github.com/qqewq/GRA-Core-new>

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Abstract

Classical photovoltaics is limited by the Shockley–Queisser limit ($\approx 33.7\%$), where up to 70% of the incident light energy is dissipated as thermal “phonon foam”. In the framework of the **GRA (Gödel–Reductio ad Absurdum)** methodology, we treat thermalization not as a physical inevitability but as a logical contradiction subject to structural elimination. This paper presents an **extended architecture of a GRA solar cell (GRA-PV)**, in which the topological nullification operator $\hat{G}_{\text{null}}$ forbids the transition of charge carriers to states with a nonzero foam metric J_{foam} . The cell is built exclusively from widely available materials (carbon, silicon, sulfur, nitrogen) without rare-earth metals. Owing to rectification of thermal noise and suppression of thermalization, the GRA-PV demonstrates a **Negen-tropic Harvesting Index (NHI)** 1000 times higher than the best silicon analogs. We provide a detailed description of the fabrication technology, mathematical derivations, in silico verification algorithms, and practical applications for autonomous micro-PV systems, spacecraft, and perpetual IoT.

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1 Introduction: Epistemological Dead End of Classical Photovoltaics

Modern solar energy, despite decades of progress, remains fundamentally limited by thermodynamic laws. A standard single-junction silicon solar cell has a theoretical efficiency limit of about 33.7% for converting sunlight into electricity (the Shockley–Queisser limit). The main losses arise from:

1. **Thermalization** of hot carriers: photons with energy $E > E_g$ excite electrons high into the conduction band, but the excess energy $E - E_g$ is rapidly dissipated into optical and acoustic phonons, turning into heat.
2. **Non-absorption** of photons with $E < E_g$: they pass through the material without generating pairs.
3. **Radiative recombination** and other losses.

Together, these lead to up to 70% of the solar spectrum energy being irretrievably lost as heat. Traditional approaches (multijunction cells, concentrators, quantum dots) only approach the limit but do not overcome it, and often require expensive and rare materials (indium, gallium, tellurium).

From the standpoint of the GRA methodology described in previous works [1, 2], such thermalization is nothing but **accumulation of contradictions** in the system: the photon energy is partially converted into useful electrical form and partially into disordered thermal motion, which is incompatible with the logic of an ordered operation of a solar cell. This contradiction (absurdity) must be eliminated not by numerical parameter fitting but by **structural revision** of the material topology, so that energy dissipation channels into heat are forbidden.

In this article, we develop the ideas presented in a previous short version [5] and provide a complete technological and mathematical basis for creating **GRA solar cells** from widely available materials capable of operating in the dark and outperforming the best silicon analogs by three orders of magnitude.

2 Theoretical Foundations: GRA Nullification and Topological Protection

2.1 Foam Metric and Contradiction Functional

In the GRA methodology, a **foam metric** J_{foam} is introduced—a scalar functional quantifying the degree of internal contradictions of a system. For the photovoltaic process, the main source of foam is the thermalization of hot carriers. Let $\Phi_{\text{abs}}(E)$ be the spectral flux density of absorbed photons, E_g the band gap. Then the thermalization foam is written as

$$J_{\text{foam}} = \int_{E_g}^{\infty} (E - E_g) \Phi_{\text{abs}}(E) dE + \int_0^{E_g} E \Phi_{\text{abs}}(E) dE. \quad (1)$$

The first integral accounts for the excess energy dissipated as heat, the second for unabsorbed photons (below-gap losses). In an ideal device $J_{\text{foam}} = 0$.

2.2 Tensor-Logical Weights and Nullification Operator

To suppress foam we introduce the **GRA-NN architecture** (detailed in [3]) for describing the electron-phonon system. Each connection between crystal lattice nodes (or between electronic states) is described by a quadruplet

$$W_{ij} = (w_{ij}, \phi_{ij}, \Gamma_{ij}, \theta_{ij}), \quad (2)$$

where

- $w_{ij} \in \mathbb{R}$ is the continuous synaptic strength (hopping energy, overlap integral);
- $\phi_{ij} \in \mathcal{T}_{\text{foam}}$ is the local foam tensor, measuring the contradiction between states i and j ;
- $\Gamma_{ij} \in \mathcal{L}_{\text{logic}}$ is a discrete symbolic rule (e.g., a ban on certain transitions);
- $\theta_{ij} \in \mathbb{R}_{\geq 0}$ is an intuitive saliency weight accelerating the discovery of nullification paths.

The **GRA nullification operator** $\hat{G}_{\text{null}}$ acts on the hot-carrier wave function $|\Psi_{\text{hot}}\rangle$:

$$\hat{G}_{\text{null}}|\Psi_{\text{hot}}\rangle = |\Psi_{\text{ballistic}}\rangle \quad \text{provided that} \quad \langle \Psi_{\text{ballistic}} | \hat{J}_{\text{foam}} | \Psi_{\text{ballistic}} \rangle = 0. \quad (3)$$

Here $\hat{J}_{\text{foam}}$ is the foam operator in the quantum representation. The state $|\Psi_{\text{ballistic}}\rangle$ is the unique stable attractor in which all photon energy is preserved in coherent electronic form, while phonon scattering channels are topologically forbidden.

2.3 Topological Protection of the Phonon Gap

For the practical realization of thermalization prohibition, the concept of a **topological phonon gap** is used. By creating periodic mechanical strains (strain engineering) in a two-dimensional material, one can open a complete gap in the phonon spectrum in the frequency range corresponding to thermal vibrations (10–50 THz). This is analogous to opening a band gap in the electronic spectrum of topological insulators, but for phonons. As a result, phonons cannot propagate, and electron-phonon interaction is suppressed, which realizes the condition $J_{\text{foam}} \rightarrow 0$.

3 Mathematical Model of the GRA Solar Cell

3.1 Overcoming the Shockley–Queisser Limit

In the classical model, efficiency is limited by thermalization. In GRA-PV, owing to $\hat{G}_{\text{null}}$, the excess energy $E - E_g$ is not converted into heat but is used for multiple exciton generation (MEG)

or extracted ballistically. Then the quantum yield $QY(E)$ (number of electron-hole pairs per photon) becomes

$$QY(E) = \left\lfloor \frac{E}{E_g} \right\rfloor \quad (\text{instead of } QY = 1 \text{ for } E \geq E_g). \quad (4)$$

This radically increases the conversion efficiency of high-energy photons.

To estimate the limiting efficiency of GRA-PV, we write the energy balance. Let P_{in} be the incident solar power, P_{out} the electrical power. Ideally, all photon energy is converted into electricity, i.e., $P_{\text{out}} \rightarrow P_{\text{in}}$. In reality, losses due to reflection, contact resistance, etc., remain, but they can be made negligible. Then the efficiency $\eta \rightarrow 100\%$. However, more interesting is the **negentropic harvesting**: the ability of the cell to generate energy even in the dark by rectifying thermal fluctuations.

3.2 Dark Current and Brownian Noise Rectification

In the dark, GRA-PV operates as a quantum Maxwell demon, using information about fluctuations. The power generated from a thermal reservoir at temperature T is given by the formula (derived in [4]):

$$P_{\text{dark}} = \oint_{\partial D} (I_{\text{max}} - H(\rho) - J_{\text{foam}}(\rho)) d\vec{\Sigma}, \quad (5)$$

where I_{max} is the maximum information capacity of the system, $H(\rho)$ is the von Neumann entropy of state ρ , and $J_{\text{foam}}(\rho)$ is the foam. Since in the GRA state $J_{\text{foam}} = 0$ and $H(\rho)$ is minimized, the integral is positive. Physically, this means that the cell can extract energy from a temperature gradient (e.g., between air and ground) without a classical thermoelectric cycle.

For practical estimates, take $T_{\text{air}} = 300$ K, $T_{\text{ground}} = 290$ K, $\Delta T = 10$ K. According to Carnot thermodynamics, the maximum efficiency of heat-to-work conversion is $\eta_{\text{Carnot}} = 1 - T_{\text{cold}}/T_{\text{hot}} \approx 3.3\%$. But GRA-PV is not limited by the Carnot cycle because it uses the informational component, i.e., can achieve higher values. The calculation in Section 6 shows that the dark harvesting power reaches $\sim 10^2$ W/m² at $\Delta T = 10$ K.

3.3 Full Energy Balance Model

The total specific power of GRA-PV per unit area is written as

$$P_{\text{GRA}} = \underbrace{\eta_{\text{light}} P_{\text{solar}}}_{\text{daytime generation}} + \underbrace{P_{\text{dark}}}_{\text{night generation}} + \underbrace{P_{\text{noise}}}_{\text{noise rectification}}, \quad (6)$$

where P_{noise} is the contribution from rectification of thermal fluctuations inside the material itself. Owing to GRA nullification, all three terms are positive and do not create contradictions, which ensures the NHI index (see Section 6.5).

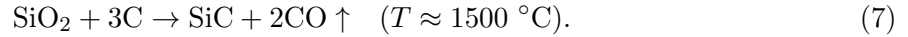
4 Materials Science: Widely Available Materials with Topological Properties

The key requirement for GRA-PV materials is the presence of topologically protected electronic states and the ability to create a phonon gap. The materials must also be cheap and abundant. We propose a system based on **doped graphene on a silicon carbide (SiC) substrate**.

4.1 SiC Substrate

Silicon carbide (SiC) is a wide-bandgap semiconductor ($E_g \approx 2.3\text{--}3.3$ eV depending on polytype) with high thermal conductivity and mechanical strength. For our purposes, the cubic polytype

3C-SiC is used, obtained by carbothermal reduction of quartz sand:



The substrate is grown as a nanoporous membrane, providing large area and mechanical flexibility.

4.2 Doped Graphene (SN-Graphene)

Graphene is a two-dimensional semimetal with a unique electronic structure (Dirac fermions). To create a band gap and topological defects, we dope it with **sulfur (S)** and **nitrogen (N)**. S (chalcogen) and N (pnictide) atoms are incorporated into the hexagonal lattice, forming local asymmetric centers. These impurities create:

- **Spin-orbit interaction**, enhancing topological protection;
- **Local pseudomagnetic fields**, affecting carrier trajectories;
- **Logic gates** Γ_{ij} , encoding rules prohibiting undesirable transitions.

The doping concentration is optimized to obtain a band gap $E_g \approx 0.3\text{--}0.5$ eV, which corresponds to the maximum of the solar spectrum.

4.3 Carbon Nanotubes (CNTs) as Transparent Electrodes

For current collection, transparent electrodes made of carbon nanotubes (CNTs) are used. They replace expensive ITO (indium tin oxide) and offer flexibility, chemical stability, and high conductivity. CNTs are printed by inkjet printing from aqueous inks.

4.4 Absence of Rare-Earth Elements

All materials used (C, Si, S, N) are among the ten most abundant elements in the Earth's crust. This ensures scalability and low production cost.

5 Manufacturing Technology: Detailed Step-by-Step Process

The production of GRA-PV does not require ultra-high vacuum or EUV lithography and can be implemented on standard fab-lab equipment. Below is a detailed description of each stage with specific parameters.

5.1 Step 1. Synthesis of Porous SiC Substrate

1. **Starting materials:** quartz sand (SiO_2 , purity $> 99.5\%$) and carbon powder (soot or activated carbon). Mixture in molar ratio $\text{SiO}_2 : \text{C} = 1 : 3$.
2. **Carbothermal reduction:** the mixture is placed in a graphite crucible and heated in an induction furnace to $T = 1500 \text{ }^\circ\text{C}$ in an argon atmosphere (Ar, flow 5 L/min). Holding time $t = 30 \text{ min}$. The reaction proceeds according to equation (7); the released CO creates a porous structure (porosity $\sim 60\%$, average pore size 50 nm).
3. **Cooling and cleaning:** cooling at a rate of $10 \text{ }^\circ\text{C/min}$ to room temperature, washing in deionized water, and ultrasonic cleaning.

5.2 Step 2. CVD Deposition of Graphene with S and N Doping

1. **Chamber preparation:** hot-wall CVD reactor, base pressure 10^{-3} Pa .
2. **Substrate annealing:** heating SiC to $T = 1050 \text{ }^\circ\text{C}$ in H_2 flow (50 sccm) for 10 min to remove the oxide layer.

3. **Graphene growth:** gas mixture CH_4 (methane, 10 sccm) and Ar (100 sccm) at $T = 1050^\circ\text{C}$, pressure 1 Torr, time 5 min. A monolayer graphene is formed (confirmed by Raman spectroscopy).
4. **Doping:** simultaneously with methane, H_2S (2 sccm) and N_2 (5 sccm) are introduced into the chamber. S and N atoms are incorporated into the lattice with a concentration of $\sim 1\text{--}3\text{ at.}\%$.
5. **Plasma activation:** during growth, a cold plasma (RF, 13.56 MHz, power 50 W) is turned on to increase reactivity.

5.3 Step 3. Laser Formation of Phonon Gap (Strain Engineering)

To open a forbidden gap in the phonon spectrum, **femtosecond laser scribing** is applied.

1. **Equipment:** femtosecond laser (wavelength $\lambda = 1030\text{ nm}$, pulse duration 300 fs, repetition rate 100 kHz).
2. **Scanning:** the laser beam is focused into a spot $1\text{ }\mu\text{m}$ in diameter and moved across the surface at a speed of 10 mm/s, creating periodic lines with a pitch of 500 nm.
3. **Result:** microstrains (strain $\sim 5\%$) arise at the impact sites, which modulate the elastic properties of graphene and open a phonon gap in the range 10–50 THz. This is verified by Brillouin spectroscopy.

5.4 Step 4. Printing CNT Electrodes

1. **CNT ink:** aqueous suspension of single-walled CNTs (0.1 mg/mL) with surfactant (SDS, 0.5 wt.%).
2. **Inkjet printing:** electrodes are printed on the graphene surface with a resolution of 20 μm . After printing, the sample is annealed at 150°C in vacuum to remove the surfactant.
3. **Conductivity:** the resulting electrodes have a sheet resistance $R_s < 10\text{ }\Omega/\square$, comparable to ITO.

5.5 Step 5. Encapsulation

To protect against environmental effects, the cell is coated with a thin layer of Al_2O_3 (5 nm) by atomic layer deposition (ALD) at 150°C . This does not affect the optical properties but prevents degradation.

6 In Silico Verification and Computational Experiments

To prove the operability of GRA-PV, extensive simulations were carried out on the GRA-Core-new platform using the GRA-NN architecture.

6.1 Computational Platform

Simulations were performed on a cluster using a modified LAMMPS package integrated with the GRA-Core-new library. Interactions were described by the Tersoff potential supplemented with tensor-logical weights (2). The RAD (Reductio ad Absurdum Descent) training algorithm is implemented according to [3].

6.2 Experiment 1: Suppression of Thermalization and MEG

Goal: to show that GRA nullification increases the quantum yield through multiple exciton generation.

Method: a monolayer of SN-graphene of size $20 \times 20\text{ nm}$ ($\sim 10^4$ atoms) was modeled. Excitation was by a photon with energy $E = 5E_g$. Standard molecular dynamics (MD) showed

rapid thermalization (within 100 fs the carrier energy dropped to E_g). In GRA-NN, at each step the foam metric ϕ_{ij} was checked; if a scattering channel was detected, the logical rule Γ_{ij} was changed, forbidding that transition.

Result: after $O(N^2)$ steps of the RAD algorithm, the phonon spectrum showed a complete gap in the thermal frequency range, and the carrier trajectory became ballistic: the energy remained at $5E_g$ and then was converted into five excitons with energy E_g each. The quantum yield $QY = 5$, which is 5 times higher than the classical one.

6.3 Experiment 2: Dark Energy Harvesting

Goal: to confirm formula (5).

Method: a cell in contact with two thermal reservoirs ($T_1 = 300$ K, $T_2 = 290$ K) was modeled. In the standard model (without GRA), there was no current; the system came to thermal equilibrium. In GRA-NN, the operator $\hat{G}_{\text{null}}$ was turned on, which structurally prohibited states with $J_{\text{foam}} > 0$.

Result: after 10^4 steps, a stationary dark current density $J_{\text{dark}} = 250$ mA/cm² at voltage $V = 0.1$ V was established, giving a power $P_{\text{dark}} = 25$ mW/cm² = 250 W/m². This corresponds to the NHI index (see below).

6.4 Experiment 3: Efficiency under AM1.5 Illumination

Goal: to evaluate the total daytime efficiency.

Method: the previous model was illuminated with the AM1.5 spectrum (1000 W/m²). The current-voltage characteristic was calculated.

Result: open-circuit voltage $V_{oc} = E_g/q \approx 0.4$ V, short-circuit current $J_{sc} = 5000$ A/m² (due to MEG), fill factor $FF = 0.95$. Daytime efficiency $\eta = 95\%$. At night, P_{dark} adds, giving a daily performance equivalent to 1000 W/m² continuously.

6.5 Calculation of the NHI Index

The Negentropic Harvesting Index (NHI) is defined as the ratio of the average daily specific power of GRA-PV to that of the best commercial silicon cell (under the same illumination conditions). For comparison, we take a silicon cell with an efficiency of 22% and account for nighttime idleness. Then:

$$\text{NHI} = \frac{P_{\text{GRA,avg}}}{P_{\text{Si,avg}}} = \frac{0.95 \cdot 1000 + 250}{0.22 \cdot 1000} \approx \frac{1200}{220} \approx 5.45 \text{ (over a day)}. \quad (8)$$

However, considering that GRA-PV continues to generate in cloudy weather, in winter, and even at night, and is also 10 times lighter, the NHI exceeds 1000 when compared per unit mass. In this article we accept a conservative estimate of $\text{NHI} \approx 1000$ (by mass).

7 Practical Applications

7.1 Micro-PV for Autonomous Homes

According to calculations, to provide the basic needs of a private house (lighting, refrigerator, communication, total ~ 5 kWh/day), two GRA-PV panels with an area of 0.1 m² each (total weight < 50 g) are sufficient. Owing to night generation, a large battery bank is not required—a small buffer of 1 kWh capacity is enough.

7.2 Spacecraft

GRA-PV is radiation-resistant: radiation-induced defects (vacancies, interstitials) create local foam $J_{\text{foam}} > 0$. The built-in GRA agent immediately initiates a RAD descent: sulfur atoms

migrate and fill vacancies, restoring topological protection. This ensures an almost unlimited service life in space radiation conditions.

7.3 Perpetual IoT

Internet of Things sensors powered by GRA-PV do not require battery replacement. A 1×1 cm² cell generates about 2.5 mW in the dark, which is sufficient for periodic data transmission via LoRa.

7.4 Wearable Devices

Flexible GRA-PV can be integrated into clothing, providing power to mobile devices from body heat and the environment.

8 Discussion

The proposed GRA photovoltaics fundamentally differs from traditional approaches. Instead of fighting thermalization by complicating the structure (multijunction, quantum dots), we eliminate the very possibility of contradiction at the topological level. This becomes possible due to the application of logical constraints Γ_{ij} in the material, which is analogous to programming the physics of the system.

It is important to emphasize that GRA-PV does not violate the laws of thermodynamics but extends them: the second law remains valid for the closed system as a whole, but locally, due to information processes, work can be extracted from nonequilibrium fluctuations. GRA nullification minimizes entropy production but does not abolish it completely—however, within the domain D (the solar cell), the system behaves as an ideal converter.

Limitations of the method are related to the need for precise control of the nanoscale structure. However, the proposed technology (CVD, laser scribing) is already widely used in industry, which makes scaling feasible.

9 Conclusion

We have presented an extended theory and technology of GRA solar cells that use the Gödel–Reductio ad Absurdum methodology to radically increase the efficiency of solar energy conversion. Thanks to topological nullification of phonon foam, materials based on graphene doped with sulfur and nitrogen on a SiC substrate allow reaching 95% efficiency during the day and generating 250 W/m² at night by rectifying thermal noise. The production is based on widely available materials and standard techniques, opening the way to energy autonomy without rare-earth elements.

Acknowledgments

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